

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-inde  
xer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-  
a3e72d95ae2ccecd5.jsonl`  
- SHA-256: `48a7475df2c59219d884c94044d8cc64beaee5ecef0788a5e72b654a0cd800c`  
- Source modified: `2026-06-16T03:26:28+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:25:35.995Z)

Adversarial Claim Verifier (voter 3/3)

Be SKEPTICAL. Try to REFUTE this claim. 2/3 refutations kill it.

Research question

Research question: **What concrete, recent (2022-2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 - ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96-99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval+serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best **recent** methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

Claim under review

"ActionFormer achieves 71.0% mAP at tIoU=0.5 on THUMOS14, outperforming the prior best model by 14.1 absolute percentage points, establishing per-moment classification + boundary regression as the state-of-the-art TAL design at the time."

****Source:**** https://dl.acm.org/doi/10.1007/978-3-031-19772-7_29 (primary)

****Supporting quote:**** "ActionFormer achieves 71.0% mAP at tIoU=0.5 on THUMOS14, outperforming the best prior model by 14.1 absolute percentage points."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:25:38.501Z)

I'll verify this claim about ActionFormer's THUMOS14 performance. Let me start by loading the search tools and checking the primary source.

3. assistant (2026-06-16T03:25:43.271Z)

I'll verify the specific numbers in this claim. The claim makes precise quantitative assertions: 71.0% mAP@0.5 on THUMOS14, and "14.1 absolute percentage points" over the prior best. Let me check the primary source and cross-reference.

4. user (2026-06-16T03:25:48.342Z)

Based on the abstract provided, ActionFormer reports **71.0% mAP at tIoU=0.5** on THUMOS14. The paper states it "outperform[s] the best prior model by 14.1 absolute percentage points."

However, the abstract does not specify whether this 14.1 percentage point improvement refers to the tIoU=0.5 metric or an average mAP score. The exact quote is:

"ActionFormer achieves 71.0% mAP at tIoU=0.5 on THUMOS14, outperforming the best prior model by 14.1 absolute percentage points."

The improvement claim is associated with the 71.0% tIoU=0.5 figure, but the abstract doesn't provide a separate average mAP number for THUMOS14.

5. user (2026-06-16T03:25:48.507Z)

Web search results for query: "ActionFormer THUMOS14 71.0 mAP tIoU 0.5 average mAP"

Links: [{"title": "ActionFormer: Localizing Moments of Actions with Transformers | Springer Nature Link", "url": "https://link.springer.com/content/pdf/10.1007/978-3-031-19772-7_29"}, {"title": "[2202.07925] ActionFormer: Localizing Moments of Actions with Transformers", "url": "https://arxiv.org/abs/2202.07925"}, {"title": "GitHub - BharathChalla/ActionFormer: ActionFormer (ECCV 2022) · GitHub", "url": "https://github.com/BharathChalla/ActionFormer"}, {"title": "GitHub - yafeng19/actionformer: Code release for ActionFormer (ECCV 2022)", "url": "https://github.com/yafeng19/actionformer"}, {"title": "GitHub - chrisindris/actionformer_release: Code release for ActionFormer (ECCV 2022) · GitHub", "url": "https://github.com/chrisindris/actionformer_release"}, {"title": "GitHub - happyharrycn/actionformer_release: Code release for ActionFormer (ECCV 2022) · GitHub", "url": "https://github.com/happyharrycn/actionformer_release"}, {"title": "actionformer_release/README.md at main · happyharrycn/actionformer_release", "url": "https://github.com/happyharrycn/actionformer_release/blob/main/README.md"}, {"title": "GitHub - csh415/actionformer_performance", "url": "https://github.com/csh415/actionformer_performance"}, {"title": "ActionFormer: Localizing Moments of Actions with Transformers", "url": "https://arxiv.org/pdf/2202.07925"}, {"title": "(PDF) ActionFormer: Localizing Moments of Actions with Transformers", "url": "https://www.researchgate.net/publication/358655346_ActionFormer_Localizing_Moments_of_Actions_with_Transformers"}]

Based on the search results, I found comprehensive information about ActionFormer's performance on THUMOS14:

ActionFormer achieves 71.0% mAP at tIoU = 0.5 on THUMOS14, outperforming the best prior model by 14.1 absolute percentage points. Additionally, the method achieves an average mAP of 66.8% ([0.3 : 0.1 : 0.7]), with an mAP of 43.9% at tIoU=0.7.

ActionFormer also demonstrates strong results on ActivityNet 1.3 (36.6% average mAP) and EPIC-Kitchens 100 (+13.5% average mAP over prior works). ActionFormer is a simple yet powerful model to identify actions in time and recognize their categories in a single shot, without using action proposals or relying on pre-defined anchor windows.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-16T03:25:57.394Z)

The numbers (71.0% mAP@0.5, 14.1 pp improvement) are confirmed verbatim from the abstract. Now let me apply the adversarial lens. The claim adds an interpretive layer beyond the raw numbers: that this "establishes per-moment classification + boundary regression as the state-of-the-art

TAL design at the time." I need to verify (a) whether the 14.1pp is correctly attributed to tIoU=0.5, and (b) whether the SOTA characterization holds. Let me check the actual paper body and a reliable third-party benchmark to confirm the 14.1 attribution and the design description.

7. user (2026-06-16T03:26:04.428Z)

ActionFormer THUMOS14 Performance

mAP Results

According to the benchmark table provided, ActionFormer achieves:

- **70.95% mAP at tIoU=0.5**
- **66.83% average mAP** across IoU thresholds (0.3-0.7)

The introduction states the model "achieves 71.0% mAP at tIoU=0.5 on THUMOS14, outperforming the best prior model by 14.1 absolute percentage points."

Design Approach

Yes, the README confirms the per-moment classification strategy: ActionFormer "classifies every moment in an input video, and regresses their corresponding action boundaries." The model performs "temporal action localization ? detecting the onsets and offsets of action instances and recognizing their action categories" in a single-shot manner without proposals or predefined anchors.

Full THUMOS14 Results Table

Metric	0.3	0.4	0.5	0.6	0.7	Avg
ActionFormer	82.13	77.80	70.95	59.40	43.87	66.83

8. user (2026-06-16T03:26:04.818Z)

Web search results for query: "ActionFormer ECCV 2022 anchor-free single-shot classification regression boundary state-of-the-art THUMOS14"

Links: [{"title": "ActionFormer: Localizing Moments of Actions with Transformers | Springer Nature", "url": "https://link.springer.com/chapter/10.1007/978-3-031-19772-7_29"}, {"title": "GitHub - yafeng19/actionformer: Code release for ActionFormer (ECCV 2022)", "url": "https://github.com/yafeng19/actionformer"}, {"title": "GitHub - BharathChalla/ActionFormer: ActionFormer (ECCV 2022) .", "url": "https://github.com/BharathChalla/ActionFormer"}, {"title": "GitHub - chrisindris/actionformer_release: Code release for ActionFormer (ECCV 2022) .", "url": "https://github.com/chrisindris/actionformer_release"}, {"title": "GitHub - happyharrycn/actionformer_release: Code release for ActionFormer (ECCV 2022) .", "url": "https://github.com/happyharrycn/actionformer_release"}, {"title": "[2202.07925] ActionFormer: Localizing Moments of Actions with Transformers", "url": "https://arxiv.org/abs/2202.07925"}, {"title": "ActionFormer: Transformer-based Action Localization", "url": "https://www.emergentmind.com/papers/2202.07925"}, {"title": "ActionFormer: Localizing Moments of Actions with Transformers", "url": "https://www.ecva.net/papers/eccv_2022/papers_ECCV/papers/136640485.pdf"}, {"title": "Revisiting Anchor Mechanisms for Temporal Action Localization", "url": "https://arxiv.org/pdf/2008.09837"}]

I've completed the web search for ActionFormer. Here are the key findings:

ActionFormer Overview

ActionFormer is one of the first Transformer-based models for temporal action localization?detecting the onsets and offsets of action instances and recognizing their action categories.

Key Technical Features

The model is trained using standard classification and regression loss, and can localize moments of actions in a single shot, without using action proposals or pre-defined anchor windows. The method adapts local self-attention to model temporal context in untrimmed videos, classifies every moment in an input video, and regresses their corresponding action boundaries.

State-of-the-Art Results

Without bells and whistles, ActionFormer achieves 71.0% mAP at tIoU=0.5 on THUMOS14, outperforming the best prior model by 14.1 absolute percentage points and crossing the 60% mAP for the first time. ActionFormer also demonstrates strong results on ActivityNet 1.3 (36.56% average mAP) and the more challenging EPIC-Kitchens 100 (+13.5% average mAP over prior works).

Publication

The paper was accepted to ECCV 2022.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-06-16T03:26:16.539Z)

All elements of the claim are now corroborated from multiple sources:

1. ****71.0% mAP@tIoU=0.5****: Confirmed (70.95% rounds to 71.0) from the official repo's benchmark table, the arXiv abstract, and the ECCV/Springer paper.
2. ****14.1 absolute percentage points over prior best****: Confirmed verbatim from the abstract and corroborated by the official README. The phrasing in the abstract directly attaches the 14.1pp improvement to the 71.0% mAP@tIoU=0.5 figure ("achieves 71.0% mAP at tIoU=0.5 ... outperforming the best prior model by 14.1 absolute percentage points"). So the attribution to tIoU=0.5 is correct, not a misread.
3. ****Design = per-moment classification + boundary regression****: Confirmed ? the paper/README explicitly describes it as classifying "every moment in an input video, and regresses their corresponding action boundaries," single-shot, anchor-free.
4. ****SOTA "at the time"*****: The abstract's own framing ("outperforming the best prior model... crossing the 60% mAP for the first time") supports the SOTA characterization for ECCV 2022. The claim is appropriately time-qualified ("at the time"), which guards against the obvious refutation that TriDet/TemporalMaxer/later models surpassed it afterward.

Source quality is high: ECCV 2022 peer-reviewed paper (primary), official author repository, and arXiv all agree. The numbers match within rounding (70.95 ? 71.0).

One mild caveat: "establishing per-moment classification + boundary regression as THE state-of-the-art TAL design" is a slight interpretive flourish ? ActionFormer demonstrated SOTA *results*, and the design family (anchor-free, per-timestep) did become dominant, but "establishing the design as SOTA" is an editorial characterization rather than a direct paper quote. However, it's a defensible and well-supported interpretation, not an overreach that would invalidate the load-bearing factual content (the numbers and the design description), which are all accurate.

This claim survives adversarial scrutiny. The quantitative core is verified verbatim against the primary source and independently corroborated by the official repository benchmark table.

10. user (2026-06-16T03:26:24.000Z)

Structured output provided successfully

11. assistant (2026-06-16T03:26:28.645Z)

I've completed the verification. The claim is ****not refuted**** ? it survives adversarial scrutiny with high confidence.

The quantitative core (71.0% mAP@tIoU=0.5 on THUMOS14, +14.1 absolute pp over prior best) is confirmed verbatim against the primary ECCV 2022 paper and arXiv abstract, and independently corroborated by the official author repository's benchmark table (70.95% @ 0.5, which rounds to 71.0). The 14.1pp improvement is correctly attributed to tIoU=0.5 (not a misread of average mAP), and the "per-moment classification + boundary regression" design description matches the paper's own language. The "at the time" qualifier appropriately scopes the SOTA claim. The only minor caveat is the editorial flourish "establishing... as the SOTA design," but it doesn't undermine any load-bearing fact. Result delivered via StructuredOutput.